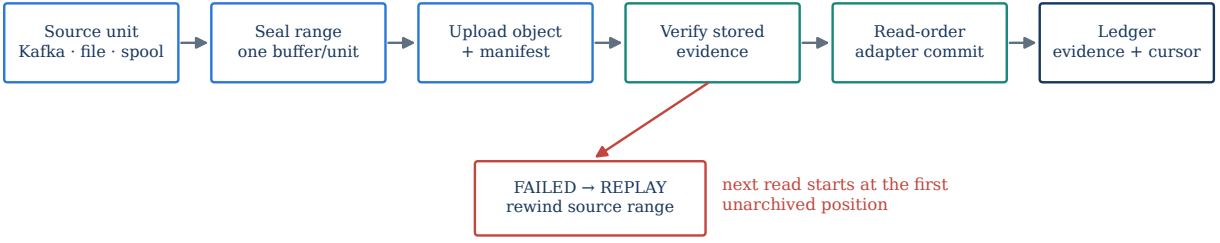
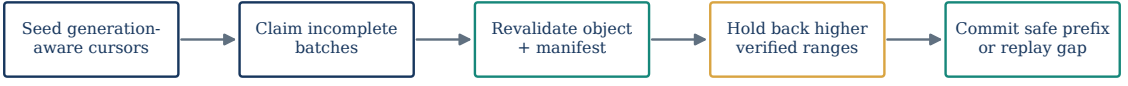
Verification-bound normal path



Crash recovery



Precondition: one in-flight batch per source unit per cycle.

At-least-once objects: replay may create duplicates; the ledger holds complete ranges.

Recovery revalidates fresh evidence before committing.